Lab: Creating Visualizations with Calculations and Calculated Fields

Author: Kevin Hanegan, Jill Jones

In this lab, you will learn how to create custom visualizations using calculations and calculated fields in Tableau.

Required Software

You will complete this lab using the free, downloadable **Tableau Desktop Public Edition** desktop software. You may also use the **Tableau Public online platform**, but some tasks may not work properly in the web-based Tableau Public environment.

Prerequisites

You need the Tableau Desktop software installed on your PC or Mac or have it already downloaded, or you can sign up for an account using the online Tableau Public platform.

Data Set

The lab uses a sample data set called **Global Sustainable Energy** and a dashboard the Tableau Public used in their tab exercises.

Download the Global Sustainable Energy data set from the following file:

https://s3.amazonaws.com/tableau-lab

Objectives

The instructor in the lab will show you how to create simple visualizations using calculations and calculated fields in Tableau.

In the lab, you will:

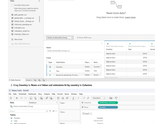
- Create custom calculations using calculations and calculated fields

Exercise: Create visualizations using calculations and calculated fields

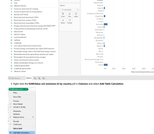
In this exercise, in Tableau Public, you will use the **Global Sustainable Energy** data set to illustrate some advanced dashboard capabilities in Tableau.

1. Open the Global Sustainable Energy data set in Tableau Public and create a new sheet.
   - Open **Global Sustainable Energy** in Tableau Desktop Public Edition and open **Sheet 1**.

2. Drag **Country** to **Rows** and **Value (all extensions for by country)** to **Columns**.



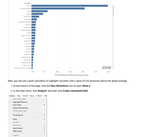
3. Expand data **WHERE(Value) and extensions for by country** and drag **Sub Table Calculations**.



4. In the **Table Calculation** window, in **Calculation Type**, select **Percent of Total**, then close the **Table Calculations** dialog box.

5. In the worksheet tab, double click the **Sheet 1** title and rename it to **CO2 Emissions by Country**, and press Enter.

6. In the table, hover to the right of the **Land Country** till you see a pale gray line. Hit the right-click to select **Sort descending**.

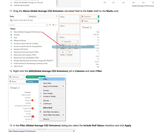


7. The table will now show the countries with the highest percentage of CO2 emissions per capita at the top of the list.

Next, you will use a quick calculation to highlight countries with a value of 50 emissions above the global average.

8. In the worksheet of the page, click the **Show Me/Analyze** icon to open **Show Me**.
   - Select the **Text** icon, then **Analysis**, and then **Global Calculated Field**.

9. Click **OK** to save the new calculated field.
10. Drag the **Global Average CO2 Emissions** field to the **Columns** **Calculated Field**.
    - Rename the new field **Global Average CO2 Emissions**.
    - Enter the formula: **[Value (all extensions for by country)] / [WHERE(Value) and extensions for by country]**.



11. Click **OK** to save the new calculated field.
12. Drag **Country** to **Rows** and drag the **Global Average CO2 Emissions** calculated field to **Columns**.

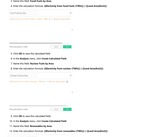
13. Drag the **Value Global Average CO2 Emissions** calculated field to the **Color** shelf in the **Marks** card.

14. Right-click the **Global Average CO2 Emissions** pill in **Columns** and select **Filter**.

15. In the **Filter Global Average CO2 Emissions** dialog box, select the **Include Null Values** checkbox and click **Apply**.

16. Deselect the **Include Null Values** checkbox and click **Apply** again. This will remove any null values in the table.

17. Click **OK**.
18. In the **Marks Global Average CO2 Emissions** legend in the right hand pane, click the down arrow and select **Edit Colors**.

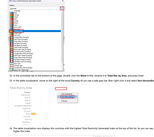


19. In the left side of the dialog box, select **Rainy** and then **Jewel Green** in the right side of the dialog box, and click **OK**.

20. In the **Marks Global Average CO2 Emissions** legend in the right pane,
    - In the worksheet tab at the bottom of the page, double click the **Sheet 2** title, rename it to **Global Average CO2 Emissions**, and press Enter.

21. Scroll down and move over the **United States** to see its value.

22. This visualization's job highlights countries with CO2 emissions that are above the global average.
    - In the main menu, click **File** and **Save Tableau Public**.


Global_Sustainable_Energy
File
Save
Save As...
Save Tableau Public
Save Tableau Public As...
Save Tableau Public Workbook As...
Save Tableau Public Data As...
Save Tableau Public Workbook Data As...
Save Tableau Public Data Source As...
Save Tableau Public Workbook Data Source As...
Save Tableau Public Data Source Connection As...
Save Tableau Public Workbook Data Source Connection As...
Save Tableau Public Data Source Connection Set As...
Save Tableau Public Workbook Data Source Connection Set As...
Save Tableau Public Data Source Connection Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set As...
Save Tableau Public Data Source Connection Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Workbook Data Source Connection Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set Set As...
Save Tableau Public Data Source Connection Set